

## CHAP2 - ELECTROSTATIC FIELD

[ Electric charge; Coulomb's law; Electric field due to point and continuous charge distribution (line, ring, disc); Electric flux; Gauss's law and its differential form; Applications of Gauss's law for spherically symmetric charge distribution; Line integral and curl of an electric field; potential and potential difference; Relation between electric field and potential; Equation of electrostatic potentials (Poisson's equation, Laplace's equation); Energy of point and continuous charge distribution; Energy density in electric field. ]

Electrostatic refers to electric charges that are stationary or "at rest," rather than flowing in a steady current. It deals with the buildup of electrical charges on the surface of objects and the attractive or repulsive forces between them.

Charges:

- a) Two types of charges viz. positive (+) and negative (-)
- b) Charge is quantized i.e  $q = ne$ , where  $n$  is an integer and  $e$  is an electronic charge.
- c) Similar charges repel and dissimilar charges attract each other.

### Coulomb's Law in Electrostatic:

Let  $Q$  be the source charge and  $q$  be the test charge at a distance  $r$  apart, the force between them is

$$\vec{F} = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^2} \hat{r} = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^3} \vec{r}$$

Let  $k = \frac{1}{4\pi\epsilon_0}$  then

$$\vec{F} = \frac{kQq}{r^3} \vec{r}$$

First lets consider these charges in free space, the force between charges be  $F_A$

$$\vec{F}_A = \frac{1}{4\pi\epsilon_0} \frac{Qq}{r^3} \vec{r}, \text{-----(1)}$$

where  $\epsilon_0 = 8.85 \times 10^{-12} \frac{C^2}{Nm^3}$  is the permittivity of free space

Now let's put these charges in a medium (like water, plastics) then the force between is  $F_m$

$$\vec{F}_m = \frac{1}{4\pi\epsilon} \frac{Qq}{r^3} \vec{r}, \text{-----(2)}$$

where  $\epsilon$  is the permittivity of the medium.

Divide equation (1) by (2)

$\vec{F}_A / \vec{F}_m = \epsilon / \epsilon_0 = \epsilon_r$ , where  $\epsilon_r$  is the relative permittivity or dielectric constant and it is always greater than 1 i.e.  $\epsilon_r > 1$

$\epsilon_r$  measures how much electrostatic force is reduced when placed in medium.

$$\epsilon_r = 1 \text{ (Vacuum)}$$

$$\epsilon_r = 1.0005 \text{ (Air)}$$

$$\epsilon_r = 11.6 \text{ (Silicon)}$$

$$\epsilon_r = 2 - 4 \text{ (Plastics)}$$

$$\epsilon_r = 78.4 \text{ (Water)}$$

$$\epsilon_r = 1000s \text{ (Ceramics)}$$

### Uses:

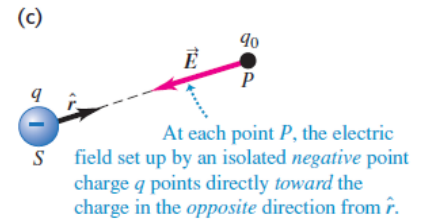
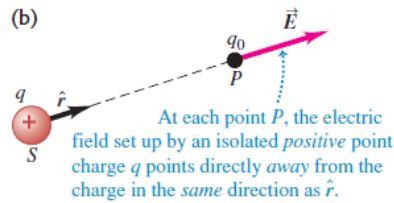
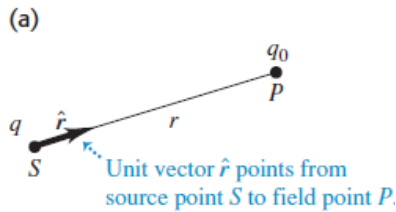
**Engineering:** It is an important parameter in the design of capacitors, printed circuit boards (PCB). Higher value means the device can be made much smaller while retaining the same energy storage capacity.

**Telecommunication:** It determines the speed of EM wave and signal propagation through cables and wireless medium.

## Electric Field or Intensity: ( $\vec{E}$ )

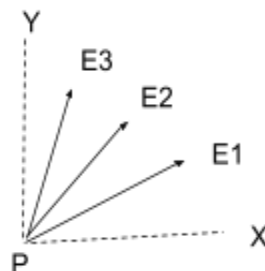
Electric field or electric intensity at a point is the force experienced by +1 Coulomb charge due to the source charge. Let  $q$  be the source charge and the intensity at a distance  $r$  from source is given by:

$$\vec{E} = \frac{Kq}{r^3} \vec{r} \text{ and its unit is Newton/Coulomb (NC}^{-1}\text{)}$$



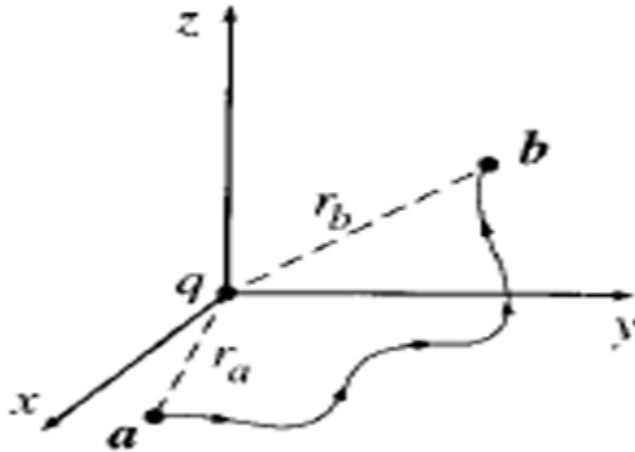
### Problems on Intensity

- a) Two charges +5C and -20C are placed at two corners of an equilateral triangle of side 1m. Find the intensity at the third corner. (Ans:  $1.62 \times 10^{10} \text{N}$ )
- b) Find the resultant intensity at  $P$ .  $\vec{E}_1, \vec{E}_2$ , and  $\vec{E}_3$  make angles  $\theta_1, \theta_2, \theta_3$  with x-axis.



## Line Integral of Electric Field

Infinitesimal displacement  $\vec{dl} = \hat{r} dr + \hat{\theta} r d\theta + \hat{\phi} r \sin\theta d\phi$



$$\int_a^b \vec{E} \cdot d\vec{l} = \int_a^b \left( \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \right) \cdot \left( dr \hat{r} + r d\theta \hat{\theta} + r \sin\theta d\phi \hat{\phi} \right)$$

$$= \frac{q}{4\pi\epsilon_0} \int_a^b \frac{1}{r^2} dr = \frac{q}{4\pi\epsilon_0} \left[ -\frac{1}{r} \right]_{r_a}^{r_b}$$

$$\boxed{= \frac{q}{4\pi\epsilon_0} \left[ \frac{1}{r_a} - \frac{1}{r_b} \right]}$$

The amount of work done by electric field from point  $a$  to  $b$  is

$$W_{ab} = \frac{q}{4\pi\epsilon_0} \left[ \frac{1}{r_a} - \frac{1}{r_b} \right]$$

Similarly, the work done by electric field from  $b$  to  $a$  is:

$$W_{ba} = \frac{q}{4\pi\epsilon_0} \left[ \frac{1}{r_b} - \frac{1}{r_a} \right]$$

Now, net work done around the closed loop is

$$W = W_{ab} + W_{ba} = 0$$

Hence the electric field is a conservative force since the line integral around the loop is zero. So,

$$\oint_c \vec{E} \cdot d\vec{l} = 0$$

Using **Stokes theorem** (converts line integral to surface integral)

$$\oint_c \vec{E} \cdot d\vec{l} = \int_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{a}$$

$$\Rightarrow \int_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{a} = 0$$

Hence  $\vec{\nabla} \times \vec{E} = 0$  for conservative field.

### **Alternately:**

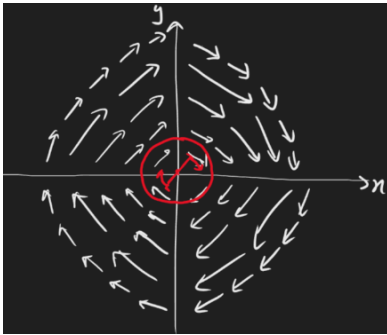
The electric field at point  $\vec{r}$  due to a point charge  $q$  located at the origin is

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{q}{r^3} \vec{r}$$

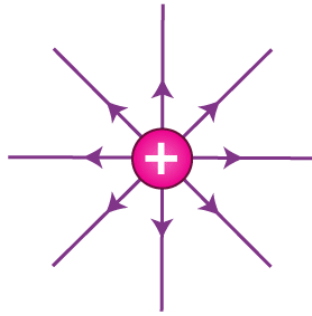
$$\begin{aligned} \text{So, } \nabla \times \vec{E} &= \nabla \times \left[ \frac{1}{4\pi\epsilon_0} \frac{q}{r^3} \vec{r} \right] = \frac{q}{4\pi\epsilon_0} \left[ \nabla \times \left( \frac{\vec{r}}{r^3} \right) \right] = \frac{q}{4\pi\epsilon_0} \left[ \frac{1}{r^3} (\nabla \times \vec{r}) + \nabla \left( \frac{1}{r^3} \right) \times \vec{r} \right] \\ &= \frac{q}{4\pi\epsilon_0} \left[ 0 + (-3r^{-3-2}) \vec{r} \times \vec{r} \right] \\ &= 0 \end{aligned}$$

What does  $\vec{\nabla} \times \vec{E} = 0$  mean?

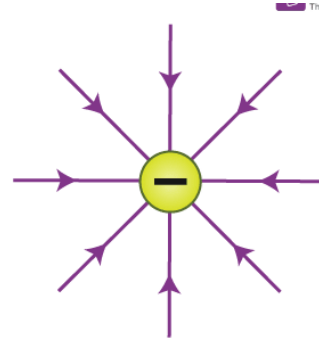
- a)  $\vec{E}$  is conservative
- b)  $\vec{E}$  field generated by a stationary charge does not “swirl” or rotate.  
We know field lines from a stationary charge are radially outward or inward without circulation.



*Static electric field does not rotate like this*



*Positive Divergence*

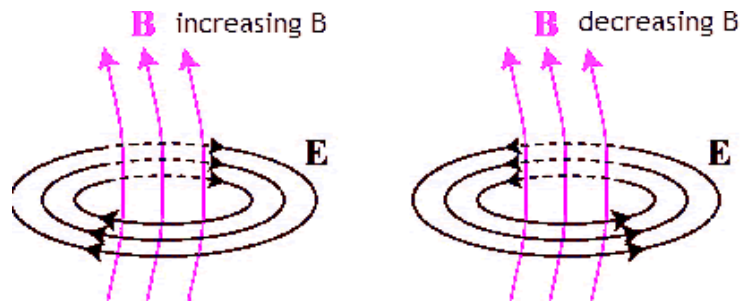


*Negative divergence*

**BUT**  $\vec{\nabla} \times \vec{E} \neq 0$  in **Electrodynamics** where time varying magnetic field produces non zero curl of  $\vec{E}$ . This means the time varying magnetic field induces **non-conservative** electric field that curls around a closed loop. This is the fundamental principle behind electrical generators and transformers.

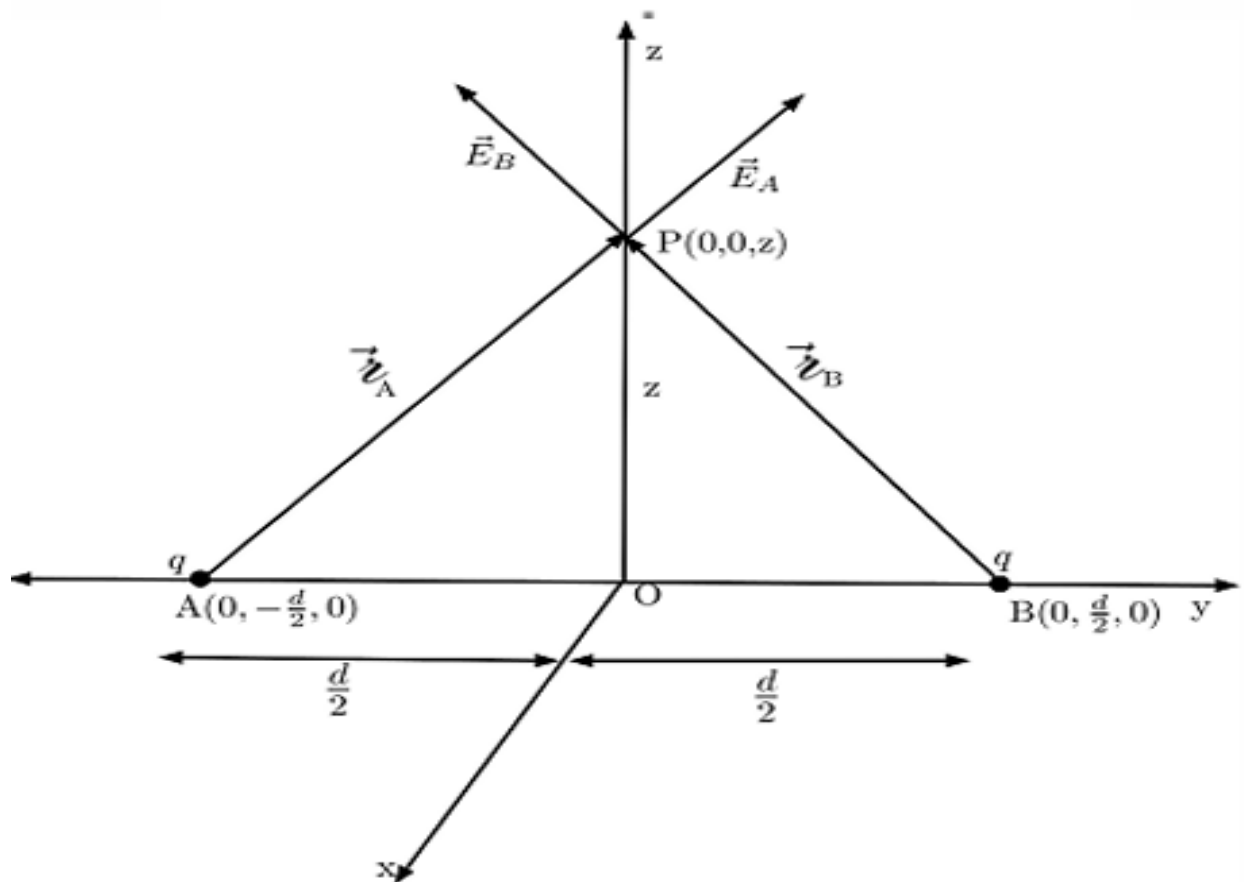
$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} \quad (\text{Maxwell Faraday Equation})$$

- a) If the magnetic field is fluctuating the electric field will “swirl” spatially.
- b) Unlike static electric fields (which converge to negative or diverge from positive charge), the electric field induced by a changing magnetic field forms a closed loop.



- c) The minus sign represents Lenz's law. It means an induced electric field creates a current whose own magnetic field opposes the original change in magnetic flux.

**I. Electric Field** a distance  $z$  above the midpoint between two equal charges,  $q$ , a distance  $d$  apart



From Figure

$$\vec{r}_A = (0-0)\hat{i} + \left(0 + \frac{d}{2}\right)\hat{j} + (z-0)\hat{k} = \frac{d}{2}\hat{j} + z\hat{k}$$

$$\Rightarrow r_A = \left(\frac{d^2}{4} + z^2\right)^{\frac{1}{2}}$$

and

$$\vec{r}_B = (0-0)\hat{i} + \left(0 - \frac{d}{2}\right)\hat{j} + (z-0)\hat{k} = -\frac{d}{2}\hat{j} + z\hat{k}$$

$$\Rightarrow r_B = \left(\frac{d^2}{4} + z^2\right)^{\frac{1}{2}}$$

- **Electric Field** at  $P$  due to the charge at A is

$$\vec{E}_A = \frac{1}{4\pi\epsilon_0} \frac{q}{r_A^3} \vec{r}_A = \frac{1}{4\pi\epsilon_0} \frac{q}{\left(\frac{d^2}{4} + z^2\right)^{\frac{3}{2}}} \left(\frac{d}{2}\hat{j} + z\hat{k}\right)$$

.....(1)

- **Electric Field** at  $P$  due to the charge at B is

$$\vec{E}_B = \frac{1}{4\pi\epsilon_0} \frac{q}{r_B^3} \vec{r}_B = \frac{1}{4\pi\epsilon_0} \frac{q}{\left(\frac{d^2}{4} + z^2\right)^{\frac{3}{2}}} \left(-\frac{d}{2} \hat{j} + z \hat{k}\right)$$

.....(2)

Therefore, **the total electric field at  $P$ :**

$$\begin{aligned} \vec{E} &= \vec{E}_A + \vec{E}_B \\ &= \frac{1}{4\pi\epsilon_0} \frac{q}{\left(\frac{d^2}{4} + z^2\right)^{\frac{3}{2}}} \left(\frac{d}{2} \hat{j} + z \hat{k}\right) + \frac{1}{4\pi\epsilon_0} \frac{q}{\left(\frac{d^2}{4} + z^2\right)^{\frac{3}{2}}} \left(-\frac{d}{2} \hat{j} + z \hat{k}\right) \end{aligned}$$

$$\therefore \boxed{\vec{E} = \frac{1}{4\pi\epsilon_0} q \frac{2z}{\left(\frac{d^2}{4} + z^2\right)^{\frac{3}{2}}} \hat{k}}$$

when  $z \gg d$

$$\text{Electric field, } \vec{E} = \frac{1}{4\pi\epsilon_0} q \frac{2z}{(z^2)^{\frac{3}{2}}} \hat{k}$$

$$\therefore \boxed{\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{2q}{z^2} \hat{k}}$$

This is the electric field at  $P$  due to  $2q$  charge at the midpoint joining  $q$  and  $q$ .

When  $z \ll d$ ,

$$\vec{E} = 0$$

## Continuous Charge and Infinitesimal elements:

### Linear Charge Density

$$\lambda = \frac{dq}{dl'}$$



charge-per-unit-length

$[dl' \rightarrow \text{an element of length along the line}]$

### Surface Charge Density

$$\sigma = \frac{dq}{da'}$$



charge-per-unit-area

$[da' \rightarrow \text{an element of area on the surface}]$

### Volume Charge Density

$$\rho = \frac{dq}{d\tau'}$$



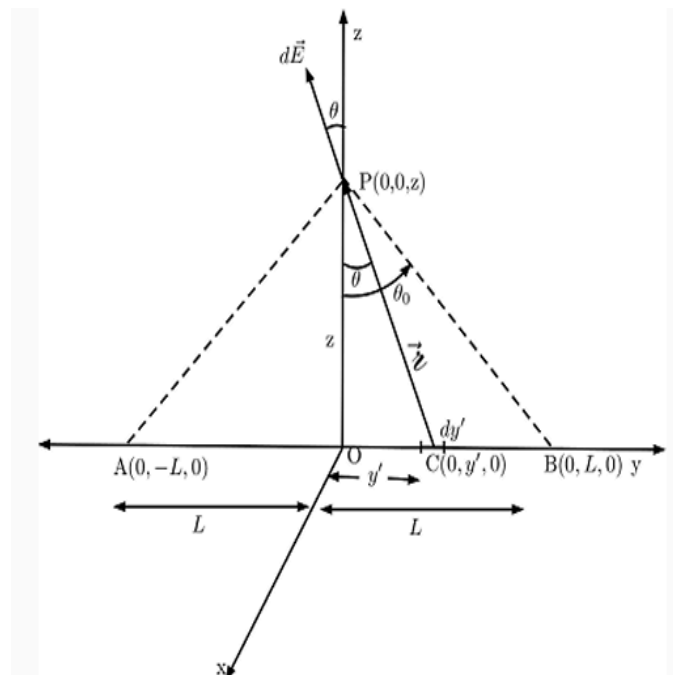
charge-per-unit-volume

$[d\tau' \rightarrow \text{an element of volume}]$

$$\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^2} \hat{i}$$



**2. Electric Field** a distance  $z$  above the midpoint of a straight line segment of length  $2L$ , which carries a uniform charge  $\lambda$



From Figure

$$\vec{r} = (0 - 0)\hat{i} + (0 - y')\hat{j} + (z - 0)\hat{k} = -y'\hat{j} + z\hat{k}$$

$$\Rightarrow r = (y'^2 + z^2)^{\frac{1}{2}}$$

The charge on an element of length  $dy'$  at C along the line is

$$\boxed{dq = \lambda dy'}$$

The electric field at  $P$  due to the charge  $dq (= \lambda dy')$  is given by

$$d\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{dq}{r^3} \vec{r} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dy'}{(y'^2 + z^2)^{\frac{3}{2}}} (-y'\hat{j} + z\hat{k})$$

.....(1)

Total Electric Field at P due to the charge on the whole line segment AB is

$$\vec{E} = \int_{-L}^{+L} d\vec{E} = \frac{1}{4\pi\epsilon_0} \lambda \left[ \int_{-L}^L \frac{(-y'\hat{j} + z\hat{k}) dy'}{(y'^2 + z^2)^{\frac{3}{2}}} \right]$$

$$\frac{1}{4\pi\epsilon_0} \lambda \left[ \int_{-L}^L \frac{-y'\hat{j} dy'}{(y'^2 + z^2)^{\frac{3}{2}}} + \int_{-L}^L \frac{z\hat{k} dy'}{(y'^2 + z^2)^{\frac{3}{2}}} \right] = \frac{1}{4\pi\epsilon_0} \lambda \left[ 2 \int_0^L \frac{z dy'}{(y'^2 + z^2)^{\frac{3}{2}}} \right] \hat{k} \quad \text{.....(2)}$$

put

$$y' = z \tan \theta$$

$$\Rightarrow dy' = z \sec^2 \theta d\theta$$

when  $y' = 0$ , then  $\theta = 0$

when  $y' = L$ , then  $\theta = \tan^{-1}\left(\frac{L}{z}\right) = \theta_0$  (say)

$$\therefore \int_{-\alpha}^{\alpha} f(x) dx = 2 \int_0^{\alpha} f(x) dx \quad ; \text{ for even function}$$

$$= 0 \quad ; \text{ for odd function}$$

$$\begin{aligned} \therefore \vec{E} &= \frac{1}{4\pi\epsilon_0} \lambda \left[ 2z \int_0^{\theta_0} \frac{z \sec^2 \theta d\theta}{z^3 \sec^3 \theta} \right] \hat{k} \\ &= \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{z} \left[ \int_0^{\theta_0} \cos \theta d\theta \right] \hat{k} \\ &= \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{z} [\sin \theta_0] \hat{k} \\ &= \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{z} \left[ \frac{L}{\sqrt{z^2 + L^2}} \right] \hat{k} \\ \therefore \boxed{\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{\lambda(2L)}{z\sqrt{z^2 + L^2}} \hat{k}} &\dots\dots\dots(3) \end{aligned}$$

For points far from the line ( $z \gg L$ ):

$$\vec{E} \cong \frac{1}{4\pi\epsilon_0} \frac{\lambda(2L)}{z^2} \hat{k}$$

$$\therefore \boxed{\vec{E} \cong \frac{1}{4\pi\epsilon_0} \frac{q}{z^2} \hat{k}} \dots\dots\dots(4)$$

For far away the line "looks" like a point charge  $q = \lambda(2L)$ .

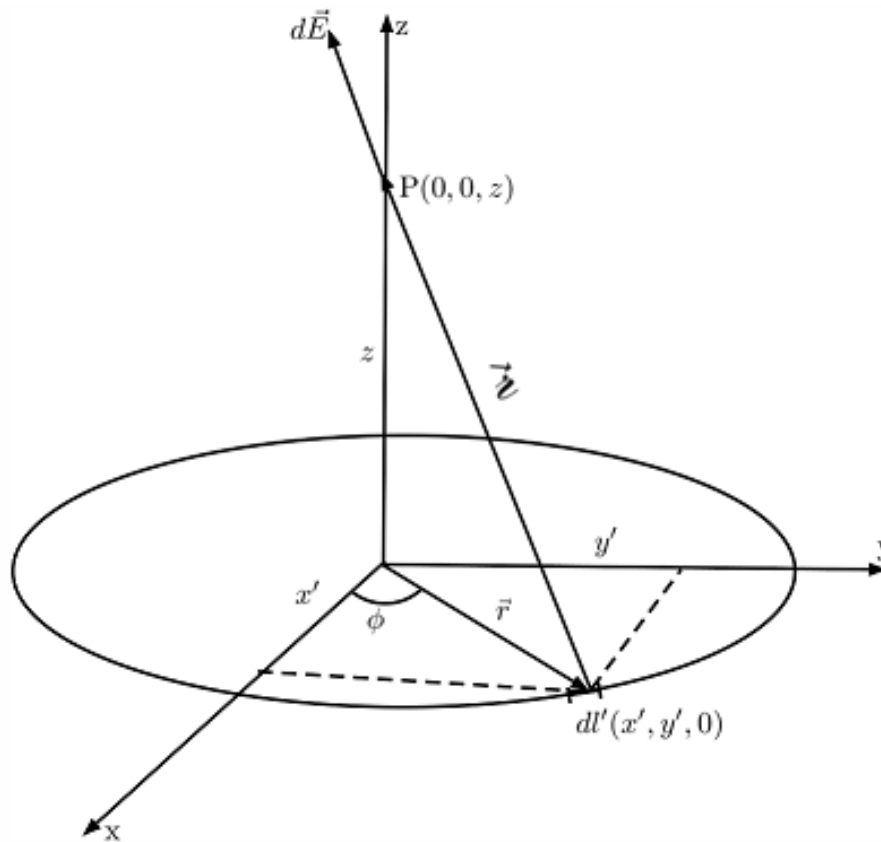
As  $L \rightarrow \infty$ :

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{\lambda(2L)}{z\sqrt{z^2 + L^2}} \hat{k} = \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{z\sqrt{\frac{z^2}{L^2} + 1}} \hat{k}$$

$$\therefore \boxed{\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{2\lambda}{z} \hat{k}} \quad [\text{Field of an infinite straight wire}]$$

\dots\dots\dots(5)

**3. Electric Field** a distance  $z$  above the centre of a circular loop of radius  $r$ , which carries a uniform line charge  $\lambda$ .



From Figure

$$\vec{r} = -x' \hat{i} - y' \hat{j} + z \hat{k} = -r \cos \phi \hat{i} - r \sin \phi \hat{j} + z \hat{k}$$

$$\text{and } r = \left( r^2 + z^2 \right)^{\frac{1}{2}}$$

The charge on an element of length

$dl'$  along a circular loop is

$$dq = \lambda dl' = \lambda (r d\phi)$$

**An elemental length  $dl'$  on the ring with coordinates  $(x', y', 0)$  subtends an elemental angle  $d\phi$  at the centre.**

The electric field at  $P$

due to the charge  $dq(= \lambda dl')$  is given by

$$\begin{aligned} d\vec{E} &= \frac{1}{4\pi\epsilon_0} \frac{dq}{r^3} \vec{r} \\ &= \frac{1}{4\pi\epsilon_0} \frac{\lambda dl'}{(r^2 + z^2)^{\frac{3}{2}}} (-r \cos \phi \hat{i} - r \sin \phi \hat{j} + z \hat{k}) \end{aligned}$$

$$\therefore \boxed{d\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{\lambda (rd\phi)}{(r^2 + z^2)^{\frac{3}{2}}} (-r \cos \phi \hat{i} - r \sin \phi \hat{j} + z \hat{k})}$$

.....(1)

$\therefore$  The net electric field at  $P$  due to the charge on whole circular loop is

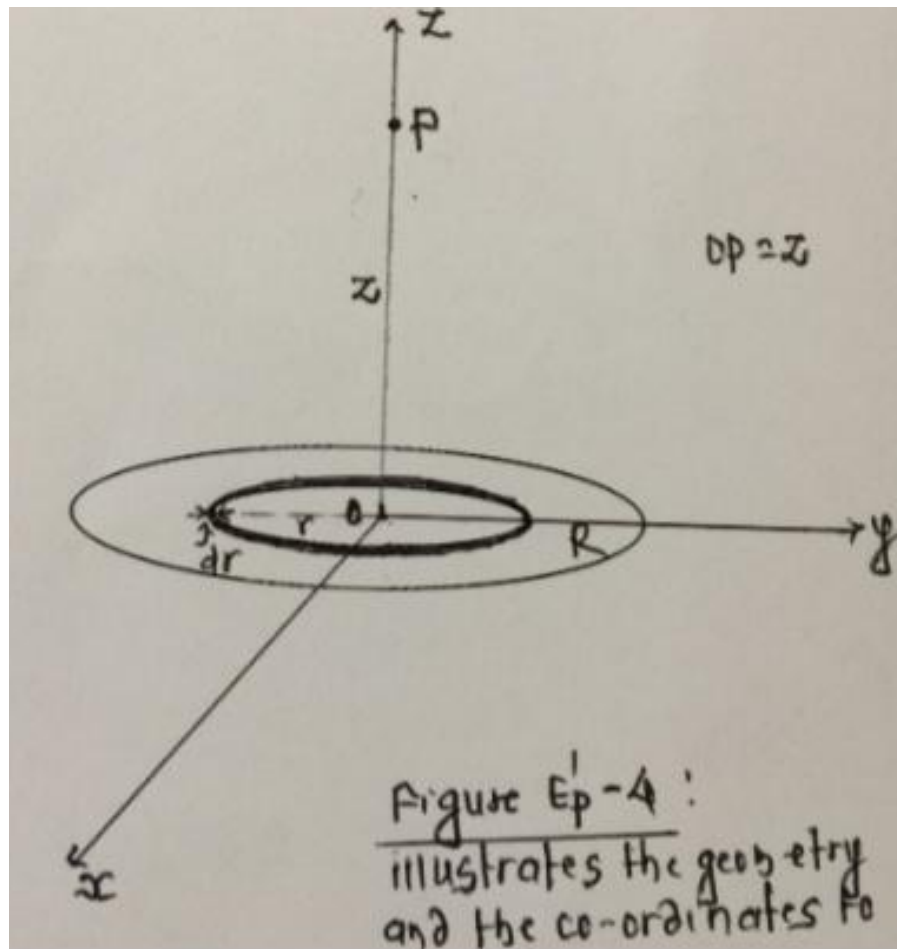
$$\begin{aligned} \vec{E} &= \int d\vec{E} \\ &= \frac{1}{4\pi\epsilon_0} \frac{r\lambda}{(r^2 + z^2)^{\frac{3}{2}}} \left[ -\left(r \int_0^{2\pi} \cos \phi d\phi\right) \hat{i} - \left(r \int_0^{2\pi} \sin \phi d\phi\right) \hat{j} + z \left(\int_0^{2\pi} d\phi\right) \hat{k} \right] \\ &= \frac{1}{4\pi\epsilon_0} \frac{r\lambda}{(r^2 + z^2)^{\frac{3}{2}}} [0 - 0 + z(2\pi) \hat{k}] \\ &= \frac{1}{4\pi\epsilon_0} \lambda (2\pi r) \frac{z}{(r^2 + z^2)^{\frac{3}{2}}} \hat{k} \end{aligned}$$

$$\therefore \boxed{\vec{E} = \frac{1}{4\pi\epsilon_0} q \frac{z}{(r^2 + z^2)^{\frac{3}{2}}} \hat{k}}$$

.....(2)

where  $q = \lambda [2\pi r]$  is the total charge  
on the circular loop.

**4. Electric Field** a distance  $z$  above the centre of a flat circular disc of radius  $R$ , which carries a uniform surface charge  $\sigma$ .



- The disk can be considered as the combination of an infinite number of infinitesimally thin rings.

Consider a ring of radius  $r$  and thickness  $dr$  of this disk.

The charge on this ring is  $dq = \sigma(2\pi r dr)$

..... (1)

The electric field at  $P$  due to the charge  $dq (= \sigma 2\pi r dr)$  on the ring is given by

$$\begin{aligned} d\vec{E} &= \frac{1}{4\pi\epsilon_0} dq \frac{z}{(r^2 + z^2)^{\frac{3}{2}}} \hat{k} \\ &= \frac{1}{4\pi\epsilon_0} (\sigma 2\pi r dr) \frac{z}{(r^2 + z^2)^{\frac{3}{2}}} \hat{k} \quad \text{using Eq.(1)} \end{aligned}$$

$$\therefore \boxed{d\vec{E} = \frac{\sigma z}{2\epsilon_0} \frac{r dr}{(r^2 + z^2)^{\frac{3}{2}}} \hat{k}} \quad \dots\dots\dots (2)$$

Total Electric field is:

$$\vec{E}_{disk} = \int d\vec{E} = \frac{\sigma z}{2\epsilon_0} \left[ \int_0^R \frac{r dr}{(r^2 + z^2)^{\frac{3}{2}}} \right] \hat{k} \quad \dots\dots\dots (3)$$

$$\begin{aligned} \text{put } r^2 + z^2 &= t^2 \\ \Rightarrow r dr &= t dt \\ \text{when } r=0, \text{ then } t &= z \\ \text{when } r=R, \text{ then } t &= \sqrt{R^2 + z^2} \end{aligned}$$

$$\begin{aligned}\therefore \vec{E} &= \frac{\sigma z}{2\epsilon_0} \left[ \int_z^{\sqrt{R^2+z^2}} \frac{t \, dt}{t^3} \right] \hat{k} = \frac{\sigma z}{2\epsilon_0} \left[ \int_z^{\sqrt{R^2+z^2}} \frac{1}{t^2} \, dt \right] \hat{k} \\ &= \frac{\sigma z}{2\epsilon_0} \left[ -\frac{1}{t} \right]_z^{\sqrt{R^2+z^2}} \hat{k} = \frac{\sigma z}{2\epsilon_0} \left[ \left( -\frac{1}{\sqrt{R^2+z^2}} \right) - \left( -\frac{1}{z} \right) \right] \hat{k}\end{aligned}$$

$$\therefore \boxed{\vec{E} = \frac{\sigma}{2\epsilon_0} \left[ 1 - \frac{z}{\sqrt{R^2+z^2}} \right] \hat{k}}$$

As  $R \rightarrow \infty$  :

$$\vec{E} = \frac{\sigma}{2\epsilon_0} [1-0] \hat{k}$$

$$= \frac{\sigma}{2\epsilon_0} \hat{k}$$

↑

Electric field due to  
infinite sheet of charge

For points far from the disk ( $z \gg R$ ):

$$\vec{E} = \frac{\sigma}{2\epsilon_0} \left[ 1 - \frac{z}{\sqrt{R^2 + z^2}} \right] \hat{k} = \frac{\sigma}{2\epsilon_0} \left[ 1 - \frac{1}{\left( \frac{R^2 + z^2}{z^2} \right)^{\frac{1}{2}}} \right] \hat{k} = \frac{\sigma}{2\epsilon_0} \left[ 1 - \left( 1 + \frac{R^2}{z^2} \right)^{-\frac{1}{2}} \right] \hat{k}$$

$$\cong \frac{\sigma}{2\epsilon_0} \left[ 1 - \left\{ 1 - \frac{1}{2} \frac{R^2}{z^2} + \dots \right\} \right] \hat{k} \quad \text{using binomial expansion}$$

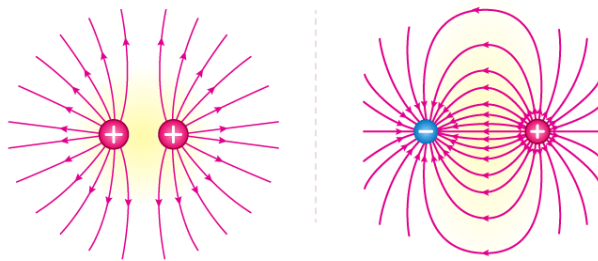
$$\cong \frac{\sigma}{2\epsilon_0} \frac{1}{2} \frac{R^2}{z^2} \hat{k} = \frac{1}{4\pi\epsilon_0} \frac{\sigma(\pi R^2)}{z^2} \hat{k}$$

$$\therefore \vec{E} \cong \frac{1}{4\pi\epsilon_0} \frac{q}{z^2} \hat{k} \quad \text{where } q = \sigma(\pi R^2) \text{ is the total charge on the disk}$$

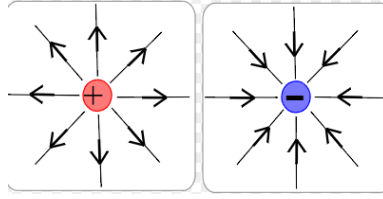
For far away the disk "looks" like a point charge  $q = \sigma(4\pi R^2)$ .

## Electric Field Lines and Flux

a) Electric field lines start from positive and end at negative.



b) Field lines due to isolated charges diverge from or converge to the charges.

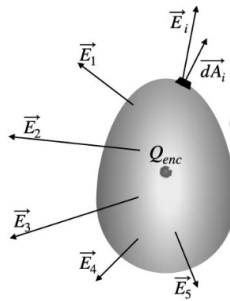


- c) The resultant electric field vector (Intensity) is tangent to the electric field line at each point.
- d) No two field lines intersect.
- e) Electric flux ( $\Phi_E$ ) is proportional to the number of electric field lines that penetrate a surface.

$$\Phi_E = \oint_S \vec{E} \cdot d\vec{A}$$

- f) The electric flux through a closed surface S measures the total charge enclosed.
- g) The SI unit of  $\Phi_E$  is  $Nm^2C^{-1}$

## Gauss Law in Electrostatic



**Statement:** The total flux  $\Phi_E$  through any closed surface is  $1/\epsilon_0$  times the charge enclosed ( $Q_{enc}$ )

$$\Phi_E = \oint_S \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$$

**Proof:** Let  $Q_{enc}$  is at a distance of  $r$  from the surface element  $d\vec{A}$ . The electric field at the surface element due to enclosed charge is,

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{Q_{enc}}{r^2} \hat{r}$$

So,

$$\oint_S \vec{E} \cdot \vec{dA} = \int \frac{1}{4\pi\epsilon_0} \frac{Q_{enc}}{r^2} \hat{r} \cdot \hat{r} r^2 \sin\theta d\theta d\phi$$

$$[\text{surface element } (\vec{dA}) = r^2 \sin\theta d\theta d\phi \hat{r}, 0 \leq \theta \leq \pi \text{ and } 0 \leq \phi \leq 2\pi]$$

$$= \int \frac{Q_{enc}}{4\pi\epsilon_0} \sin\theta d\theta d\phi$$

$$= \frac{Q_{enc}}{4\pi\epsilon_0} \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi$$

$$= \frac{Q_{enc}}{4\pi\epsilon_0} \pi \cdot (2\pi)$$

$$= \frac{Q_{enc}}{\epsilon_0}$$

$$\left[ \oint_S \vec{E} \cdot \vec{dA} = \frac{Q_{enc}}{\epsilon_0} \right]$$

If  $Q_{enc} = q_1 + q_2 + q_3 + \dots + q_n$  then,

$$\oint_S \vec{E} \cdot \vec{dA} = \frac{1}{\epsilon_0} \sum_{i=1}^n q_i$$

For continuous charge distribution with charge density  $\rho$  and volume element  $d\tau$

$$\oint_S \vec{E} \cdot \vec{dA} = \frac{1}{\epsilon_0} \oint_V \rho d\tau \text{ ----(1)}$$

Using Gauss Divergence theorem (*surface integral to volume integral*),

$$\oint_S \vec{E} \cdot d\vec{A} = \oint_V (\vec{\nabla} \cdot \vec{E}) d\tau \quad \text{---(2)}$$

From (1) and (2)

$$\Rightarrow \oint_V (\vec{\nabla} \cdot \vec{E}) d\tau = \oint_V \frac{\rho}{\epsilon_0} d\tau$$

Equating integrand

$$\Rightarrow (\vec{\nabla} \cdot \vec{E}) = \frac{\rho}{\epsilon_0} \quad (\text{First Maxwell equation})$$

**This is a differential form of Gauss law.**

### **Physical meaning:**

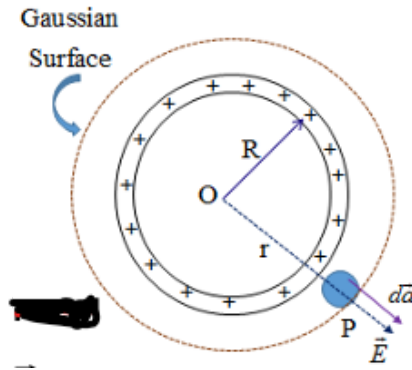
Since the divergence is not zero, it measures how much "field" is spreading outward or inward from a location. A positive divergence (for  $\rho > 0$ ) means electric field lines are originating or expanding outward, while a negative divergence (for  $\rho < 0$ ) means they are terminating or contracting inward.

### **Application of Gauss Law:**

**1. Use Gauss's Law to find the electric field (a) outside, (b) on and (c) inside a spherical shell of radius  $R$ , which carries a uniform surface charge density  $\sigma$ .**

The total charge on the spherical shell is  $4\pi R^2 \sigma = Q$  (suppose)

**a) Electric Field Outside a Spherical Shell:**



- a) P is the point of interest which is at  $r > R$ .  
 b) The charge enclosed by Gaussian sphere of radius  $r$  is

$$Q_{enc} = Q = 4\pi R^2 \sigma$$

- c) From Gauss law:

$$\oint_S \vec{E} \cdot d\vec{A} = \frac{1}{\epsilon_0} Q_{enc} \Rightarrow E \int dA = \frac{1}{\epsilon_0} Q$$

$$E \times 4\pi r^2 = \frac{1}{\epsilon_0} Q$$

$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \left( \frac{Q}{r^2} \right)$$

*The electric field outside the spherical shell is following an inverse square law.*

### **b) Electric field on the surface of spherical shell ( $r = R$ ):**

If you replace  $r$  with  $R$ , it gives you the electric field on the surface of the spherical surface of radius  $R$ .

$$\oint_S \vec{E} \cdot d\vec{A} = \frac{1}{\epsilon_0} Q_{enc} \Rightarrow E \int dA = \frac{1}{\epsilon_0} Q$$

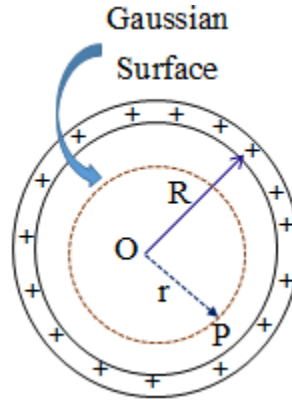
$$E \times 4\pi R^2 = \frac{1}{\epsilon_0} Q$$

$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \frac{Q}{R^2}$$

*The electric field on the surface of a spherical shell is constant*

at every point on the surface.

c) Electric field inside the surface of spherical shell ( $r < R$ ):

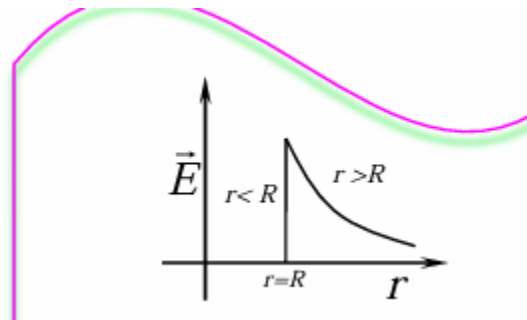


The charge enclosed by the Gaussian sphere is zero i.e.  $Q_{enc} = 0$  because the charged spherical shell lies outside of the Gaussian surface.

From Gauss law:

$$\oint_S \vec{E} \cdot d\vec{A} = \frac{1}{\epsilon_0} Q_{enc} \Rightarrow E \int dA = 0$$
$$\Rightarrow E = 0$$

The plot of electric field  $\vec{E}$  vs  $r$  from center to outside of the spherical shell is given below.

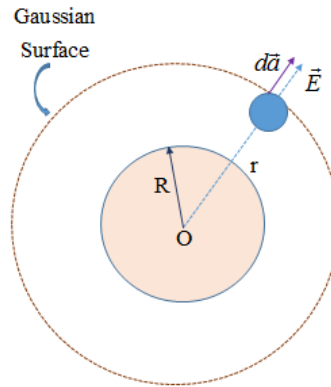


Plot of  $\vec{E}$  vs  $r$  for spherical shell carrying charge  $q$

**2. Use Gauss's Law to find the electric field (a) outside, (b) on and (c) inside a uniformly charged solid sphere of radius  $R$ , which carries a volume charge density  $\rho$ .**

The total charge on the solid sphere is  $\frac{4}{3}\pi R^3 \rho = Q$  (suppose)

a) Outside solid sphere of radius  $R$ :



The total charge on the sphere is  $Q_{enc} = \frac{4}{3}\pi R^3 \rho = Q$ . The point of interest is at  $r > R$ .

$$\oint_S \vec{E} \cdot d\vec{A} = \frac{1}{\epsilon_0} Q_{enc} \Rightarrow E \int dA = \frac{1}{\epsilon_0} Q$$

$$E \times 4\pi r^2 = \frac{1}{\epsilon_0} Q$$

$$E = \frac{Q}{4\pi\epsilon_0 r^2}$$

The electric field outside the charged solid sphere is following an inverse square law.

b) On the sphere ( $r = R$ )

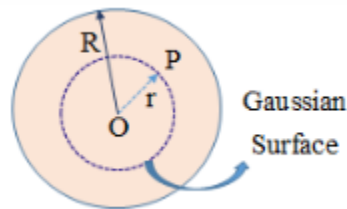
$$\oint_S \vec{E} \cdot d\vec{A} = \frac{1}{\epsilon_0} Q_{enc} \Rightarrow E \int dA = \frac{1}{\epsilon_0} Q$$

$$E \times 4\pi R^2 = \frac{1}{\epsilon_0} Q$$

$$E = \frac{Q}{4\pi\epsilon_0 R^2}$$

*This shows that the electric field on the surface of a solid sphere is constant everywhere.*

c) *Inside a Uniformly Charged solid sphere*



Let  $r$  be the radius of Gaussian sphere then the total charge enclosed by Gaussian sphere is  $Q_{enc} = \frac{4}{3}\pi r^3 \rho$

$$\oint_S \vec{E} \cdot d\vec{A} = \frac{1}{\epsilon_0} Q_{enc} \Rightarrow E \int dA = \frac{1}{\epsilon_0} Q_{enc}$$

$$E (4\pi r^2) = \frac{1}{\epsilon_0} \frac{4}{3}\pi r^3 \rho$$

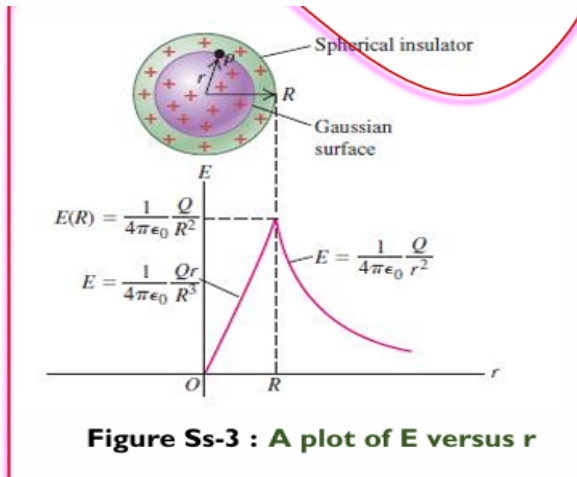
$$E = \frac{\rho r}{3\epsilon_0}$$

$$\text{But } \rho = \frac{Q}{\frac{4}{3}\pi R^3}$$

$$\Rightarrow E = \frac{Qr}{4\pi R^3 \epsilon_0}$$

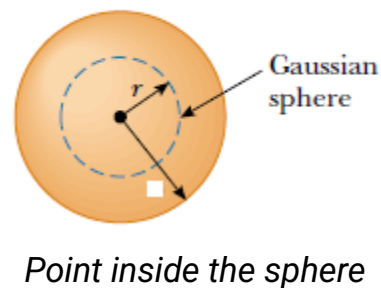
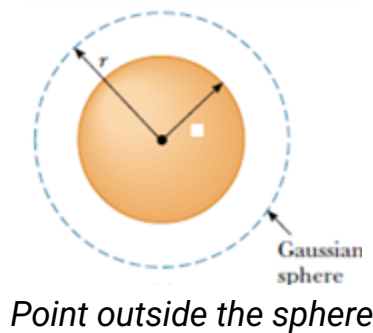
*This shows that the electric field inside the solid sphere varies with distance from the center of the sphere.  $E$  is zero at the center of the sphere.*

The plot of  $E$  vs  $r$  for a uniformly charged solid sphere is given below.



At the centre of the sphere  $r = 0$ , Electric Field is Zero.

**4. Find the electric field (a) outside (b) on and (c) inside a sphere, which carries a charge density proportional to the distance from the origin, for some constant  $k$ .**



The point of interest is at a distance  $r$  from the center of the sphere of radius  $R$ . The sphere carries a charge density proportional to distance from the center, for some constant  $k$  i.e.  $\rho = kr$

**(a) Outside sphere**

**Note:**

The volume element  $d\tau = r^2 \sin\theta \, dr \, d\theta \, d\phi$  where  
 $0 \leq r \leq R$ ;  $0 \leq \theta \leq \pi$ ;  $0 \leq \phi \leq 2\pi$

$$Q_{enc} = \int \rho \, d\tau = \int (kr) \, r^2 \sin\theta \, dr \, d\theta \, d\phi$$

$$Q_{enc} = k \int_0^R r^3 \, dr \int_0^\pi \sin\theta \, d\theta \int_0^{2\pi} d\phi$$

$$Q_{enc} = k \frac{R^4}{4} (2)(2\pi) = \pi k R^4$$

From Gauss law:

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} Q_{enc}$$

$$E (4\pi r^2) = \frac{1}{\epsilon_0} (\pi k R^4)$$

$$\Rightarrow E = \frac{k}{4\epsilon_0} \frac{R^4}{r^2} \quad (E \text{ varies as } r^2)$$

**On the surface of sphere ( $r = R$ )**

$$\Rightarrow E = \frac{k}{4\epsilon_0} R^2 \quad (E \text{ is constant and maximum on the surface})$$

**(b) Inside sphere**

$$Q_{enc} = \int \rho \, d\tau = \int (kr) \, r^2 \sin\theta \, dr \, d\theta \, d\phi$$

$$Q_{enc} = k \int_0^r r^3 \, dr \int_0^\pi \sin\theta \, d\theta \int_0^{2\pi} d\phi$$

$$Q_{enc} = k \frac{r^4}{4} (2)(2\pi) = \pi k r^4$$

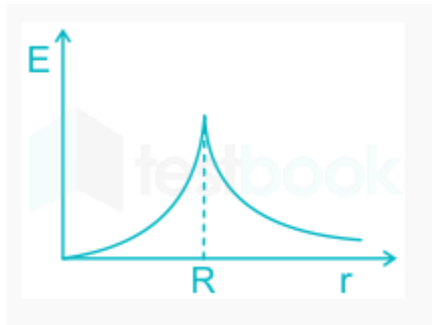
From Gauss law:

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} Q_{enc}$$

$$E (4\pi r^2) = \frac{1}{\epsilon_0} (\pi k r^4)$$

$$\Rightarrow E = \frac{k}{4\epsilon_0} r^2 \quad (E \text{ increases with } r^2 \text{ from the center to surface of sphere})$$

The behaviour of electric field



## Electric Potential

- a) Electric potential at an arbitrary point P is the potential energy per unit charge at P.

$$V = \frac{U}{q}$$

- b) This potential at point P is equal to the amount of work done to bring positive unit charge from infinity to the point.

$$V(\vec{r}) = V_P = W_{unit}(\infty \rightarrow P)$$

Let  $q$  be the source charge. The potential due to charge  $q$  at a distance  $r$  from it is given by

$$V(\vec{r}) = \int_{\infty}^r \vec{E} \cdot d\vec{r} = - \int_{\infty}^r kq/r^2 dr = kq/r, \text{ where } k = 1/(4\pi\epsilon_0)$$

- a) Potential of a point charge at P

$$V(\vec{r}) = kq/r, \text{ where } k = 1/(4\pi\epsilon_0)$$

- b) Potential of collection of charges at P

$$V(\vec{r}) = k \sum_{i=0}^n \frac{q_i}{r_i}, \text{ where } k = 1/(4\pi\epsilon_0)$$

- c) Potential due to continuous charge distribution

$$V(\vec{r}) = k \int \frac{dq}{r}, \text{ where } k = 1/(4\pi\epsilon_0)$$

- c) It is a scalar quantity and the unit is Joules/Coulomb or volt.
- d) It obeys superposition principle i.e.  $V=V_1+V_2+V_3+\dots$
- e) Potential Energy (U) at point P is the total amount of work done to carry Q amount of charge (NOT +1 Coulomb charge) from infinity to P. Thus,

$$U = W(\infty \rightarrow P) = Q \times V(\vec{r})$$

### **The Electric Field is the Gradient of a Scalar Potential**

The potential difference between two points  $a$  and  $b$ :

$$V(b) - V(a) = -\int_a^b \vec{E} \cdot d\vec{l} \quad \dots\dots\dots(1)$$

The fundamental theorem for gradients states that:

$$V(b) - V(a) = \int_a^b (\nabla V) \cdot d\vec{l} \quad \dots\dots\dots(2)$$

So, 
$$-\int_a^b \vec{E} \cdot d\vec{l} = \int_a^b (\nabla V) \cdot d\vec{l}$$

Since this is true for any points  $a$  and  $b$ , the integrands must be equal:

$$\therefore \boxed{\vec{E} = -\nabla V}$$

### **Poisson's and Laplace's Equation**

Gauss law in differential form is  $\vec{\nabla} \cdot \vec{E} = \frac{1}{\epsilon_0} \rho$ . Since  $\vec{E} = -\nabla V$ , we can

write Gauss law in differential form as

$$\vec{\nabla} \cdot \vec{E} = \nabla \cdot (-\nabla V) = \frac{1}{\epsilon_0} \rho$$

$$\Rightarrow \nabla^2 V = -\frac{1}{\epsilon_0} \rho \text{ (This is Poisson's Equation)}$$

In a region where there is no charge then  $\rho = 0$ . The Poisson's equation reduces to

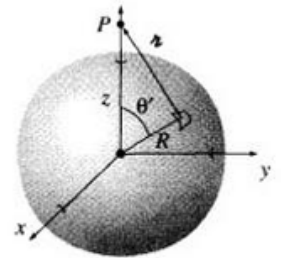
$$\nabla^2 V = 0 \text{ (This is Laplace's Equation)}$$

### I. Find the potential of a uniformly charged spherical shell of radius .

#### Solution:

- The potential for a surface charge is  $V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\sigma}{t} da'$ .

- From the law of cosines,  $t^2 = R^2 + z^2 - 2Rz \cos \theta'$



- An element of surface area on this sphere is  $R^2 \sin \theta' d\theta' d\phi'$ .

So,

$$\begin{aligned} V(z) &= \frac{1}{4\pi\epsilon_0} \left[ \int \frac{\sigma}{\sqrt{R^2 + z^2 - 2Rz \cos \theta'}} (R^2 \sin \theta' d\theta' d\phi') \right] \\ &= \frac{\sigma R^2}{4\pi\epsilon_0} \left[ \left\{ \int_0^\pi \frac{\sin \theta' d\theta'}{\sqrt{R^2 + z^2 - 2Rz \cos \theta'}} \right\} \left\{ \int_0^{2\pi} d\phi' \right\} \right] \\ &= \frac{\sigma R^2}{4\pi\epsilon_0} \left[ \left\{ \frac{1}{Rz} \left( \sqrt{(R+z)^2} - \sqrt{(R-z)^2} \right) \right\} \{2\pi\} \right] \\ &= \frac{\sigma R^2}{2\epsilon_0} \left[ \frac{1}{Rz} \left( \sqrt{(R+z)^2} - \sqrt{(R-z)^2} \right) \right] \\ \therefore V(z) &= \frac{\sigma R}{2\epsilon_0 z} \left[ \sqrt{(R+z)^2} - \sqrt{(R-z)^2} \right] \end{aligned}$$

$$\text{Put } R^2 + z^2 - 2Rz \cos \theta' = t^2$$

$$\Rightarrow \sin \theta' d\theta' = \frac{1}{Rz} (t dt)$$

$$\text{when } \theta' = 0, \text{ then } t = \sqrt{R^2 - z^2}$$

$$\text{when } \theta' = \pi, \text{ then } t = \sqrt{R^2 + z^2}$$

$$\Rightarrow \int_0^\pi \frac{\sin \theta' d\theta'}{\sqrt{R^2 + z^2 - 2Rz \cos \theta'}} = \frac{1}{Rz} \int_{\sqrt{R^2 - z^2}}^{\sqrt{R^2 + z^2}} \frac{(t dt)}{t}$$

$$= \frac{1}{Rz} \int_{\sqrt{R^2 - z^2}}^{\sqrt{R^2 + z^2}} dt$$

$$= \frac{1}{Rz} \left( \sqrt{(R+z)^2} - \sqrt{(R-z)^2} \right)$$

For points outside the sphere,  $z > R$ ,

and hence  $\sqrt{(R-z)^2} = z - R$

$$\begin{aligned}\therefore V_{out}(z) &= \frac{R\sigma}{2\epsilon_0 z} [(R+z) - (z-R)] = \frac{\sigma R^2}{\epsilon_0 z} \\ &= \frac{1}{4\pi\epsilon_0} \frac{\sigma(4\pi R^2)}{z} = \frac{1}{4\pi\epsilon_0} \frac{q}{z}\end{aligned}$$

For points inside the sphere,  $z < R$ ,

and hence  $\sqrt{(R-z)^2} = R - z$

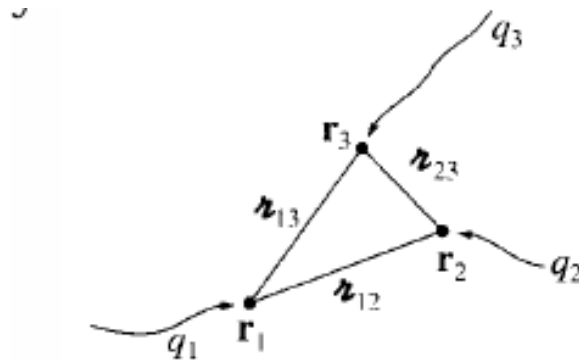
$$\begin{aligned}\therefore V_{in}(z) &= \frac{R\sigma}{2\epsilon_0 z} [(R+z) - (R-z)] = \frac{\sigma R}{\epsilon_0} \\ &= \frac{1}{4\pi\epsilon_0} \frac{\sigma(4\pi R^2)}{R} = \frac{1}{4\pi\epsilon_0} \frac{q}{R}\end{aligned}$$

For points on the sphere,  $z = R$ ,

$$\begin{aligned}\therefore V_{on}(R) &= \frac{\sigma R}{\epsilon_0} = \frac{1}{4\pi\epsilon_0} \frac{\sigma(4\pi R^2)}{R} \\ &= \frac{1}{4\pi\epsilon_0} \frac{q}{R} = V_{in}(z)\end{aligned}$$

## Work and Energy in Electrostatic

### Electric Potential Energy



- Consider that three point charges  $q_1, q_2$  and  $q_3$  are lying at locations  $\vec{r}_1, \vec{r}_2$  and  $\vec{r}_3$  respectively.
- First of all, let us remove all the three charges to infinite distance from each other.
- (i) Let us move the charge  $q_1$  from infinity to its location  $\vec{r}_1$ .  
The work done to move the charge  $q_1$  from infinity to its location  $\vec{r}_1$  is  $W_1=0$ .

(ii) Let us move the charge  $q_2$  from infinity to its location  $\vec{r}_2$ .

The work done to move the charge  $q_2$  from infinity to its location  $\vec{r}_2$  is

$$W_2 = q_2 [V_1(\vec{r}_2)] \quad \text{where } V_1(\vec{r}_2) \text{ is the potential due to } q_1.$$

$$= q_2 \left[ \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_{12}} \right] = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

(iii) Let us move the charge  $q_3$  from infinity to its location  $\vec{r}_3$ .

The work done to move the charge  $q_3$  from infinity to its location  $\vec{r}_3$  is

$$W_3 = q_3 [V_{1,2}(\vec{r}_3)] \quad \text{where } V_{1,2}(\vec{r}_3) \text{ is the potential due to charges } q_1 \text{ and } q_2.$$

$$= q_3 \left[ \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_{13}} + \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_{23}} \right] = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{r_{13}} + \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{r_{23}}$$

Therefore, total work done to assemble three charges is  $W = W_1 + W_2 + W_3$  and is equal to electric potential energy ( $U$ )

$$\therefore U = W = 0 + \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}} + \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{r_{13}} + \frac{1}{4\pi\epsilon_0} \frac{q_2 q_3}{r_{23}} = \frac{1}{4\pi\epsilon_0} \left[ \frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right]$$

$$= \frac{1}{2} \times \frac{1}{4\pi\epsilon_0} \sum_{i=1}^3 \sum_{\substack{j=1 \\ j \neq i}}^3 \frac{q_i q_j}{r_{ij}}$$

**For a system of n-point charges, we have**

$$U = \frac{1}{2} \times \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \frac{q_i q_j}{r_{ij}}$$

$$= \frac{1}{2} \sum_{i=1}^n q_i \left( \frac{1}{4\pi\epsilon_0} \sum_{\substack{j=1 \\ j \neq i}}^n \frac{q_j}{r_{ij}} \right)$$

$$\therefore U = \frac{1}{2} \sum_{i=1}^n q_i V(\vec{r}_i)$$

where  $V(\vec{r}_i) = \frac{1}{4\pi\epsilon_0} \sum_{\substack{j=1 \\ j \neq i}}^n \frac{q_j}{r_{ij}}$  is the potential at point  $\vec{r}_i$

(the position of  $q_i$ ) due to all other charges.

## The Energy of Continuous Charge Distribution

Since the total work done to assemble n-point charges is given by,

$$W = \frac{1}{2} \sum_{i=1}^n q_i V(\vec{r}_i) \text{ -----(1)}$$

where  $V(\vec{r}_i)$  is the potential at point  $\vec{r}_i$  (the position of  $q_i$ ) due to all other charges.

For continuous charge distribution, we have,

$$q = \int \rho \, d\tau$$

Equation (1) becomes

$$W = \frac{1}{2} \int \rho V(\vec{r}) d\tau \text{ -----(2)}$$

Using Gauss Law

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \Rightarrow \rho = \epsilon_0 (\vec{\nabla} \cdot \vec{E})$$

Equation (2) becomes,

$$W = \frac{1}{2} \int \epsilon_0 (\vec{\nabla} \cdot \vec{E}) V \, d\tau$$

$$\text{Or, } W = \frac{\epsilon_0}{2} \int V (\vec{\nabla} \cdot \vec{E}) \, d\tau$$

Using the relation  $\vec{\nabla} \cdot (V\vec{E}) = V(\vec{\nabla} \cdot \vec{E}) + (\vec{\nabla} V) \cdot \vec{E}$

$$W = \frac{\epsilon_0}{2} \int [\nabla \cdot (V\vec{E}) - (\nabla V) \cdot \vec{E}] d\tau$$

$$\text{Or, } W = \frac{\epsilon_0}{2} \int [\nabla \cdot (V\vec{E}) - (\nabla V) \cdot \vec{E}] d\tau$$

$$W = \frac{\epsilon_0}{2} \int [\nabla \cdot (V\vec{E}) + \vec{E} \cdot \vec{E}] d\tau \quad (\text{since } -(\nabla V) = \vec{E})$$

$$W = \frac{\epsilon_0}{2} \int \nabla \cdot (V\vec{E}) d\tau + \frac{\epsilon_0}{2} \int \vec{E} \cdot \vec{E} d\tau$$

Using Gauss Divergence theorem

$$\oint_V \nabla \cdot (V\vec{E}) d\tau = \oint_S (V\vec{E}) \cdot d\vec{a}$$

$$W = \frac{\epsilon_0}{2} \oint_S (V\vec{E}) \cdot d\vec{a} + \frac{\epsilon_0}{2} \int E^2 d\tau$$

When the integration is taken over all space, the surface integral goes to zero.

$$W = \frac{\epsilon_0}{2} \int_{\text{all space}} E^2 d\tau = \int_{\text{all space}} u_E d\tau$$

Where  $u_E = \frac{1}{2} \epsilon_0 E^2$  is called energy density (Energy per unit volume (Joules/m<sup>3</sup>))